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Education

- 2015 – 2021 📖 **Ph.D. (Dr.-Ing.)**, Max Planck Institute for Dynamics of Complex Technical Systems & Otto von Guericke University Magdeburg, Germany.
- 2013 – 2015 📖 **M.Sc.** King Abdullah University of Science and Technology, Saudi Arabia.
- 2009 – 2013 📖 **B.Eng.** Huazhong University of Science and Technology, China.

Research experience

- 2026 📖 **Visiting Professor**, Kyoto University, Japan. (Host: Prof. Ryoichi Kurose)
- 2024 📖 **Visiting Professor**, CERFACS, France. (Host: Prof. Bénédicte Cuenot)
- Since 2021 📖 **Group leader**, Otto von Guericke University Magdeburg, Germany. (Host: Prof. Dominique Thévenin)
- 2015 – 2021 📖 **Graduate Research Assistant**, Max Planck Institute for Dynamics of Complex Technical Systems & Otto von Guericke University Magdeburg, Germany. (Supervisor: Prof. Dominique Thévenin)
- 2013 – 2015 📖 **Graduate Research Assistant**, King Abdullah University of Science and Technology, Saudi Arabia. (Supervisor: Prof. Hong G. Im)
- 2012 – 2013 📖 **Undergraduate Research**, Huazhong University of Science and Technology, China.

Honors, Awards and Scholarships

- 2026 📖 **JSPS** (Japan Society for the Promotion of Science) fellowship.
- 2025 📖 **“NIC Excellence Project of 2025”** awarded to the HPC project “Dadoren” on JUWELS as PI and project manager (2 out of 180+ granted HPC projects on supercomputers in Germany).
- 2021 📖 **“summa cum laude”** for the PhD thesis (Best possible thesis grade in Germany).
- 2020 📖 **Chinese Government Award** for outstanding self-financed students abroad (the highest award to graduate students studying outside China).
- 2015 📖 **IMPRS** (International Max Planck Research School) scholarship.
- 2013 📖 **KAUST** research scholarship.
- 2012 📖 **The Lloyd’s Register Educational Trust Scholarship – Best Paper Award.**
- 2010 📖 **National Scholarship** (the highest award to undergraduates in China).

Publications (H-index: 16)

First/corresponding author Journal Articles

- 1 **C. Chi**, Z. Chen, and D. Thévenin, “Ultra-lean diffusion-controlled 2d premixed hydrogen/air flames,” *Proceedings of the Combustion Institute*, in press, 2026.
- 2 **C. Chi**, B. Cuenot, and D. Thévenin, “Turbulent flame-wall interaction: Dynamics of flame thickness and combustion regime,” *Journal of Fluid Mechanics*, vol. 1029, A5, 2026.

- 3 C. Chi, A. Dernbecher, K. P. R. Subramanian, *et al.*, "Chemical reaction neural networks for semi-detailed kinetics including the influence of inorganics and secondary reactions during cellulose pyrolysis," *Energy Conversion and Management*, vol. 348, p. 120 655, 2026.
- 4 C. Chi, W. Han, and D. Thévenin, "Local extinction characteristics of turbulent premixed h₂/air flame under sub-atmospheric pressure conditions: A dns study," *Combustion and Flame*, vol. 283, p. 114 580, 2026.
- 5 C. Chen, C. Chi, W. Han, L. Yang, and D. Thévenin, "A flame marker for ammonia/hydrogen/air premixed flames during flame/wall interactions," *Proceedings of the Combustion Institute*, vol. 41, p. 105 935, 2025.
- 6 C. Chi, W. Guan, V. Inna, and D. Thévenin, "Reactivity stratified flame to enhance ammonia combustion with hydrogen: Effects of temperature, pressure, downstream mixture, and molecular diffusion," *Proceedings of the Combustion Institute*, vol. 41, p. 105 907, 2025.
- 7 C. Chi, Z. Ou, C. Yu, W. Han, and D. Thévenin, "A generalized memory effect in fluid/flame dynamics due to unsteady events," *Particuology*, vol. 103, pp. 232–241, 2025.
- 8 C. Chi, "Flame dynamic insights into emission characteristics of NH₃/H₂/air combustion in turbulent boundary layers," *Combustion and Flame*, vol. 269, p. 113 723, 2024.
- 9 C. Chi, C. Yu, B. Cuenot, U. Maas, and D. Thévenin, "Effect of differential diffusion on head-on quenching of premixed NH₃/H₂/air flames within turbulent boundary layers," *Proceedings of the Combustion Institute*, vol. 40, p. 105 276, 2024.
- 10 W. Guan, C. Chi, W. Liang, and D. Thévenin, "Revisiting performance of reactivity stratification with hydrogen addition for ammonia combustion," *Proceedings of the Combustion Institute*, vol. 40, p. 105 381, 2024.
- 11 C. Chi, W. Guan, Z. Ou, K. Sundmacher, and D. Thévenin, "Direct numerical simulations of polypropylene gasification in supercritical water.," *Physics of Fluids*, vol. 35, no. 6, p. 063 321, 2023.
- 12 C. Chi, H. Theisel, and D. Thévenin, "Interaction of a turbulent flame with the very-large-scale structures in a channel flow.," *European Journal of Mechanics / B Fluids*, vol. 101, pp. 167–175, 2023.
- 13 C. Chi and D. Thévenin, "DNS study on reactivity stratification with prechamber H₂/air turbulent jet flame to enhance NH₃/air combustion in gas engines.," *Fuel*, vol. 347, p. 128 387, 2023.
- 14 C. Chi, A. Abdelsamie, and D. Thévenin, "Transient ignition of premixed methane/air mixtures by a pre-chamber hot jet: A DNS study," *Flow, Turbulence and Combustion*, vol. 108, no. 3, pp. 775–795, 2022.
- 15 C. Chi, W. Han, and D. Thévenin, "Effects of molecular diffusion modeling on turbulent premixed nh₃/h₂/air flames," *Proceedings of the Combustion Institute*, vol. 39, no. 2, pp. 2259–2268, 2022.
- 16 C. Chi, S. Sreekumar, and D. Thévenin, "Data-driven discovery of heat release rate markers for premixed NH₃/H₂/air flames using physics-informed machine learning," *Fuel*, vol. 330, p. 125 508, 2022.
- 17 C. Chi, D. Thévenin, A. J. Smits, S. Wolligandt, and H. Theisel, "Identification and analysis of very-large-scale turbulent motions using multiscale proper orthogonal decomposition," *Physical Review Fluids*, vol. 7, no. 8, p. 084 603, 2022.
- 18 C. Chi, X. Xu, and D. Thévenin, "Efficient premixed turbulent combustion simulations using flamelet manifold neural networks: A priori and a posteriori assessment," *Combustion and Flame*, vol. 245, p. 112 325, 2022.
- 19 C. Chi, G. Janiga, and D. Thévenin, "On-the-fly artificial neural network for chemical kinetics in direct numerical simulations of premixed combustion," *Combustion and Flame*, vol. 226, pp. 467–477, 2021.
- 20 C. Chi, A. Abdelsamie, and D. Thévenin, "A directional ghost-cell immersed boundary method for incompressible flows," *Journal of Computational Physics*, vol. 404, p. 109 122, 2020.
- 21 C. Chi, G. Janiga, K. Zähringer, and D. Thévenin, "DNS study of the optimal heat release rate marker in premixed methane flames," *Proceedings of the Combustion Institute*, vol. 37, no. 2, pp. 2363–2371, 2019.

- 22 C. Chi, A. Abdelsamie, and D. Thévenin, "Direct numerical simulations of hotspot-induced ignition in homogeneous hydrogen-air pre-mixtures and ignition spot tracking," *Flow, Turbulence and Combustion*, vol. 101, no. 1, pp. 103–121, 2018.
- 23 C. Chi, G. Janiga, A. Abdelsamie, K. Zähringer, T. Turányi, and D. Thévenin, "DNS study of the optimal chemical markers for heat release in syngas flames," *Flow, Turbulence and Combustion*, vol. 98, no. 4, pp. 1117–1132, 2017.
- 24 C. Chi, B. J. Lee, and H. G. Im, "An improved ghost-cell immersed boundary method for compressible flow simulations," *International Journal for Numerical Methods in Fluids*, vol. 83, no. 2, pp. 132–148, 2017.

Co-authored Journal Articles

- 1 W. Guan, C. Chi, A. G. Szanthoffer, and D. Thévenin, "A reduced kinetic mechanism for ammonia/hydrogen mixtures with alleviated stiffness and high accuracy tailored for high-fidelity numerical simulations," *Applications in Energy and Combustion Science*, vol. 24, p. 100 377, 2025.
- 2 W. Guan, F. Gharibi, C. Chi, A. Abdelsamie, and D. Thévenin, "A ghost-cell immersed boundary method for reacting flow simulations with conjugate heat transfer," *Journal of Computational Physics*, vol. 543, p. 114 399, 2025.
- 3 A. Hemaizia, D. Thévenin, and C. Chi, "Detached eddy simulation (des) of a turbulent premixed flame stabilized on a bluff body," *Proceedings in Applied Mathematics and Mechanics*, vol. 25, e70007, 2025.
- 4 C. Yu, C. Chi, C. Tang, and B. Gorr, "A numerical study of the stagnation flow premixed lean hydrogen/air flame stabilized at the wall with the focus on no emission and flame-solid interaction," *Journal of Thermal Science*, vol. 34, pp. 707–719, 2025.
- 5 C. Chen, C. Chi, D. Thévenin, W. Han, and L. Yang, "Effects of cryogenic temperature on turbulent premixed hydrogen/air flames," *Proceedings of the Combustion Institute*, vol. 40, p. 105 749, 2024.
- 6 Z. Ou, Y. Wan, C. Chi, Q. Xue, and D. Thévenin, "A parameter-free and monolithic approach for multi-scale simulations of flows, transport, and chemical reactions in porous media," *Journal of Computational Physics*, vol. 514, p. 113 203, 2024.
- 7 Z. Ou, Y. Wan, Q. Xue, C. Chi, F. Gharibi, and D. Thévenin, "A one-domain pore-resolved approach for multiphase flows in porous media," *Physics of Fluids*, vol. 36, p. 063 108, 2024.
- 8 Y. Wang, C. Xu, C. Chi, and Z. Chen, "Direct numerical simulations of turbulent premixed cool flames: Global and local flame dynamics analysis," *Combustion and Flame*, vol. 270, p. 113 759, 2024.
- 9 Y. Wang, C. Xu, C. Chi, Y. Yang, and Z. Chen, "Temperature effect on turbulent burning velocity of lean premixed hydrogen/air flames," *Physics of Fluids*, vol. 36, p. 125 165, 2024.
- 10 A. Abdelsamie, S. Voss, C. Chi, *et al.*, "Direct numerical simulation of the transitional airflow in a patient-specific larynx during exhalation," *Computers and Fluids*, vol. 255, p. 105 819, 2023.
- 11 W. Guan, A. Abdelsamie, C. Chi, Z. He, and D. Thévenin, "A dedicated reduced kinetic model for ammonia-/dimethyl-ether turbulent premixed flames," *Combustion and Flame*, vol. 257, p. 113 002, 2023.
- 12 S. Wolligant, C. Rössl, C. Chi, D. Thévenin, and H. Theisel, "Autonomous particles for in-situ-friendly flow map sampling," *Proceedings of the Conference on Vision, Modeling, and Visualization*, pp. 189–197, 2023.
- 13 C. Yu, L. Cai, C. Chi, S. Mashruk, A. Valera-Medina, and U. Maas, "Numerical investigation on the Head-on Quenching (HoQ) of laminar premixed lean to stoichiometric ammonia–hydrogen–air flames," *Flow, Turbulence and Combustion*, vol. 111, no. 4, pp. 1301–1322, 2023.
- 14 Z. Ou, C. Chi, L. Guo, and D. Thévenin, "A directional ghost-cell immersed boundary method for low Mach number reacting flows with interphase heat and mass transfer," *Journal of Computational Physics*, vol. 468, p. 111 447, 2022.

- 15 Z. Ou, L. Guo, **C. Chi**, J. Zhao, H. Jin, and D. Thévenin, "Fully resolved direct numerical simulation of single coal particle gasification in supercritical water," *Fuel*, vol. 329, p. 125 474, 2022.
- 16 Z. Ou, L. Guo, **C. Chi**, *et al.*, "Interface-resolved direct numerical simulations of interphase momentum, heat, and mass transfer in supercritical water gasification of coal," *Physics of Fluids*, vol. 34, no. 10, p. 103 319, 2022.
- 17 A. Abdelsamie, **C. Chi**, M. Nanjaiah, I. Skenderović, S. Suleiman, and D. Thévenin, "Direct numerical simulation of turbulent spray combustion in the SpraySyn burner: Impact of injector geometry," *Flow, Turbulence and Combustion*, vol. 106, no. 2, pp. 453–469, 2021.
- 18 Z. Zhang, A. Abdelsamie, **C. Chi**, D. Thévenin, and K. H. Luo, "Combustion mode and mixing characteristics of a reacting jet in crossflow," *Energy & Fuels*, vol. 35, no. 16, pp. 13 325–13 337, 2021.

Peer Reviewed Articles in Proceedings

- 1 **C. Chi** and D. Thévenin, "Interactions of thermodiffusive instabilities and turbulence in lean hydrogen flames: Effect of turbulence length scale," Edinburgh: [Proceedings of 12th European Combustion Meeting](#), 2025, pp. 84–89.
- 2 W. Guan, **C. Chi**, A. Abdelsamie, Z. He, and D. Thévenin, "Structure and pollutant emissions of ammonia/-dimethyl-ether/air turbulent premixed flames: A dns study," Edinburgh: [Proceedings of 12th European Combustion Meeting](#), 2025, pp. 23–28.
- 3 R. Al-Hamadi, A. Abdelsamie, **C. Chi**, G. Janiga, and D. Thévenin, "Direkte numerische simulation der pyrolyse von biomasse-partikeln," Paderborn: [Proceedings of 32. Deutscher Flammentag](#), 2025, pp. 1–7.
- 4 A. Abdelsamie, **C. Chi**, Z. Ou, and D. Thévenin, "Direct numerical simulation of gasification of a polypropylene plastic particle in supercritical water: Impact of flow conditions," Algarve: [Proceedings of 14th European Conference on Industrial Furnaces and Boilers](#), 2024, pp. 1–8.
- 5 **C. Chi**, A. Abdelsamie, W. Guan, Z. Ou, and D. Thévenin, "DNS of gasification process of polypropylene (PP) plastics in supercritical water at $Re = 100$," VDI, Ed., Berlin: [Proceedings of 31. Deutscher Flammentag](#), 2023, pp. 1–9.
- 6 **C. Chi** and D. Thévenin, "Turbulent flame dynamics and pollutant emission characteristics of stratified NH_3/H_2 /air flames in a prechamber/main chamber system: A DNS study," Rouen: [Proceedings of 11th European Combustion Meeting](#), 2023, pp. 2461–2466.
- 7 W. Guan, A. Abdelsamie, **C. Chi**, Z. He, and D. Thévenin, "A reduced chemical kinetic mechanism for ammonia/dimethyl-ether combustion," Rouen: [Proceedings of 11th European Combustion Meeting](#), 2023, pp. 2408–2413.
- 8 **C. Chi**, A. Smits, S. Wolligandt, H. Theisel, and D. Thévenin, "Identification of very-large-scale motions in turbulent channel flows by direct numerical simulations and data-driven decomposition," Stuttgart: [Proceedings of NIC Symposium](#), 2022, pp. 415–424, ISBN: 978-3-95806-646-5.
- 9 **C. Chi**, A. Abdelsamie, and D. Thévenin, "Pre-chamber turbulent hot jet ignition by DNS: Impact of turbulence," Napoli: [Proceedings of 10th European Combustion Meeting](#), 2021, pp. 421–425.
- 10 **C. Chi** and D. Thévenin, "Effect of detailed diffusion models on NO production in NH_3/H_2 /air turbulent jet flames using direct numerical simulations," VDI, Ed., Bochum: [Proceedings of 30. Deutscher Flammentag](#), 2021, pp. 143–152.
- 11 **C. Chi** and D. Thévenin, "Impact of different diffusion models on NO production in NH_3/H_2 /air turbulent flames using DNS," Xi'an: [Proceedings of Global Power and Propulsion Society GPPS21](#), 2021, GPPS-TC-2021–3, ISBN: 2504-4400.
- 12 **C. Chi**, A. Abdelsamie, and D. Thévenin, "Towards direct numerical simulations of a gas engine pre-chamber," Lisbon: [Proceedings of 9th European Combustion Meeting](#), 2019, pp. 1–7.

- 13 A. Abdelsamie, G. Janiga, **C. Chi**, and D. Thévenin, "Turbulence structure analysis of DNS data using DMD and SPOD: Mixing jet and channel flow," Italy: [Progress in Turbulence VII](#), 2017, pp. 199–204, ISBN: 978-3-319-57934-4.
- 14 **C. Chi**, A. Abdelsamie, T. Oster, and D. Thévenin, "Probability of hotspot ignition and ignition spot tracking in turbulent hydrogen-air mixtures using direct numerical simulations," Dubrovnik: [Proceedings of 8th European Combustion Meeting](#), 2017, pp. 925–930, ISBN: 978-953-59504-1-7.
- 15 **C. Chi**, G. Janiga, K. Zähringer, and D. Thévenin, "Indirekte messung der wärmefreisetzung in wasserstoffflammen mittels spezieskorrelationen," VDI, Ed., Düsseldorf: [Proceedings of 28. Deutscher Flammentag](#), 2017, pp. 625–636, ISBN: 978-3-18-102302-0.

Third-party funding

- 2024 – 2027  **DFG Priority Program 2419 (AddBluff4NH₃/H₂):** Additively manufactured Bluff body burner investigated by DNS and experiments for fuel flexible, stable and safe combustion of NH₃/H₂ mixtures
Financial support: **€645,300**
Own contribution: **Co-PI, Project Coordinator**, Supervision of Dr. Guan in the project
- 2023 – 2026  **DFG Project TH881/38-1 (DADOREN):** Direct numerical simulations And Data-driven analysis Of ignition and combustion in Realistic pre-chamber/ Engine systems with NH₃/H₂ blend fuel.
Financial support: **€265,600**
Own contribution: **Co-PI, Project Coordinator, Project manager**
- 2022 – 2024  **German Excellence Research Initiative "SmartProSys":** Project o6: Thermochemical decomposition under supercritical conditions.
Financial support: **€132,200**
Own contribution: **Person in charge**
- 2021 – 2022  **European Regional Development Fund, operation number ZS/2021/10/161306:** Direct Numerical Simulations of prechamber hot jet ignition.
Financial support: **€65,000**
Own contribution: **Person in charge**
- 2020 – 2022  **DFG Priority Program 1881 (grant number: TH881/30-1):** DNS and Visual Analysis of Superstructures in Turbulent Channels with Mixing by Parallel Injection.
Financial support: **€263,400**
Own contribution: **Proposal preparation, Project manager**
- 2018 – 2020  **European Regional Development Fund, operation number ZS/2017/01/83590, project number 1704/00084:** Reman-Gasmotor mit Vorkammerzündung.
Financial support: **€183,200**
Own contribution: **Person in charge**

HPC projects

- 2026-2028  "Direct Numerical Simulation of gasification and pyrolysis of polypropylene" on SuperMUC-NG with **24.73 Mcore-h** and **900 GPU-h**
Own contribution: **Proposal preparation, Project manager**

HPC projects (continued)

- 2026-2027 ■ "Direct Numerical Simulation of Pyrolysis of Biomass Particles" on **JUWELS** with **17 Mcore-h**
Own contribution: **Proposal preparation, Project manager**
- 2023-2027 ■ "Direct numerical simulations of ignition and combustion in realistic pre-chamber/engine systems with NH_3/H_2 blend fuel" on **JUWELS** with **55.7 Mcore-h**
Own contribution: **PI, Project initiation, Project manager**
Award: **NIC Excellence Project 2025**
- 2022-2024 ■ "Study of turbulent flame properties for ammonia blended with DME by Direct Numerical Simulation" on **SuperMUC-NG** with **10.52 Mcore-h**
Own contribution: **Proposal preparation, Project manager**
- 2021-2024 ■ "Sound Induced by Oscillating Flows in a Moving Airway" on **JUWELS** with **19.5 Mcore-h**
Own contribution: **Proposal preparation**
- 2020-2023 ■ "DNS and Visual Analysis of Superstructures in Turbulent Channels with Mixing by Parallel Injection" on **JUWELS** with **35 Mcore-h**
Own contribution: **Project initiation, Proposal preparation, Project manager**
- 2017-2022 ■ "Modulation of Turbulent Properties in a Spray Flame Burning n-Heptane: Direct Numerical Simulation" on **SuperMUC-NG** with **88.6 Mcore-h**
Own contribution: **Proposal preparation, Project manager**

Teaching activities

- Full lectureship in "Advanced Fluid Dynamics", 04/2026-07/2026
- Full lectureship in "Computational Fluid Dynamics", 10/2025-02/2026
- Full lectureship in "Advanced Fluid Dynamics", 04/2025-07/2025
- Full lectureship in "Computational Fluid Dynamics", 10/2024-02/2025
- Lectureship in "Advanced Fluid Dynamics", 04/2024-07/2024
- Teaching assistant and part lectureship in "Computational Fluid Dynamics", 04/2024-07/2024
- Teaching assistant and part lectureship in "Computational Fluid Dynamics", 10/2023-02/2024
- Teaching assistant in "Computational Fluid Dynamics", 04/2023-07/2023

Supervising activities

- 2018 ■ **Master thesis:** DNS in engine-like geometry (by Mr. Ke Li)
- 2020 ■ **Master thesis:** Artificial neural networks for describing reactive systems (by Mr. Alexey Kropman)
- 2022 ■ **Bachelor thesis:** Comparison between PCA and autoencoder for reduced manifold of premixed flames (by Mr. Pascal Bock)
- 2023 ■ **Master thesis:** Reducing detailed chemical kinetic mechanism for ammonia/hydrogen/air combustion by using artificial neural networks (by Mr. Ayachitula Yagna Murthy)
- **Master thesis** (co-supervised by **Dr. Sauerland from Hitachi**): Physics-informed neural networks for heat transfer problem (by Mr. Srijith Sreekumar)
- **Master thesis** (co-supervised by **Prof. F. Mauss from BTU**): Validation of a diesel spray simulation on experimental data (by Mr. Viswanath Reddy Saribala)

Supervising activities (continued)

- **Master thesis** (co-supervised by **Dr. B. Shu from PTB**): Laser diagnostics (TDLAS) and high-temperature gas-phase reaction kinetics for Ammonia and blend combustion (by Mr. Hariprasad Valiya Parambath)
- 2024 ■ **Master thesis**: Chemical Reaction Neural Networks (CRNN) to discover the detailed kinetics for biomass pyrolysis: effect of catalysis and diffusion (by Mr. Kesav Prasath R. S.)
- 2025 ■ **Master thesis**: Local combustion mode classification using machine learning techniques (by Mr. Peigang Yang)
- **Master thesis** (co-supervised by **Prof. R. X. Fernandes from PTB**): Parametric investigation of ammonia-hydrogen flame characteristics in a non-premixed burner (by Mr. B.K. Kodaru)
- **Master thesis**: Flame speed enhancement and NO_x reduction in reactivity stratified NH₃/H₂/air premixed flames (by Mr. Vivek Inna)
- 2026 ■ **Bachelor thesis**: Dimensionsreduktion von Flammendaten mit Autoencoder und Vergleich des latenten Raums mit einer CEMA (by Mr. Oliver Schmitz)

Journal review activities

- Regular reviewer for Journal of Fluid Mechanics, Combustion and Flame, Proceedings of the Combustion Institute, Flow Turbulence and Combustion, Physics of Fluids, Physical Review Fluids, Journal of Computational Physics, Energy, Fuel ... (100+ papers reviewed until today)

References

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|--------------------------|---|
| Prof. Dominique Thévenin | ■ Otto von Guericke University Magdeburg, Germany. thevenin@ovgu.de |
| Prof. Hong G. Im | ■ KAUST, Saudi Arabia. hong.im@kaust.edu.sa |
| Prof. Bénédicte Cuenot | ■ SAFRAN, France. benedicte.cuenot@safrangroup.com |